

Example 1. Use factorials to produce a solution to $x^2 \equiv -1 \pmod{41}$.

Solution. Since 41 is

solutions to $x^2 \equiv -1 \pmod{41}$ are given by

We regroup

$$\begin{aligned}
 & \pm(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5)(6 \cdot 7)(8 \cdot 9)(10 \cdot 11)(12 \cdot 13)(14 \cdot 15)(16 \cdot 17)(18 \cdot 19)20 \\
 & \equiv \pm(-3)(1)(-10)(-13)(-8)(5)(-15)(14)(10) \pmod{41} \\
 & \equiv \pm(3 \cdot 10 \cdot 13)(8 \cdot 5 \cdot 15)(14 \cdot 20) \pmod{41} \\
 & \equiv \pm(-20)(-15)(-7) \pmod{41} \\
 & \equiv \pm(20 \cdot 15)(7) \pmod{41} \\
 & \equiv \pm(13)(7) \equiv
 \end{aligned}$$

Example 2. Let p be a prime of the form $p = 4k + 3$. Prove that if

$a^2 + b^2 \equiv 0 \pmod{p}$, then:

Proof. Suppose towards a contradiction that there exist a and b so

Then $a \not\equiv 0 \pmod{p}$ and thus a has an inverse modulo p .

$$a^2 + b^2 \equiv 0 \pmod{p}$$

Thus $x^2 \equiv -1 \pmod{p}$ is solved by:

This is impossible, as -1 has no square roots modulo p when p is prime of form:

□

A. **Arithmetic Functions.** An **arithmetic function** is a function

For example

The arithmetic function τ is defined as follows. If n is a positive integer, then

$$\tau(n) =$$

For example

$$\tau(1) =$$

$$\tau(12) =$$

if p is prime, then

$$\tau(p) =$$

and if further k is a positive integer

$$\tau(p^k) =$$

Another arithmetic function is

The arithmetic function σ is defined as follows. If n is a positive integer, then

$$\sigma(n) =$$

For example

$$\sigma(1) =$$

$$\sigma(12) =$$

if p is prime, then

$$\sigma(p) =$$

and if further k is a positive integer

$$\sigma(p^k) =$$

and then using the geometric sum formula $1 + x + \cdots + x^n = \frac{x^{n+1} - 1}{x - 1}$ becomes

$$\sigma(p^k) =$$

B. Summation Notation. We will work with summation notation.

If f is arithmetic then

$$\sum_{d|n} f(d) =$$

For example

$$\sum_{d|12} f(d) =$$

and thus

$$= \sum_{d|12} d =$$

$$= \sum_{d|12} 1 =$$

In summation notation

$$\tau(n) =$$

$$\sigma(n) =$$

We can also use double sum notation, for example

$$\sum_{\substack{d_1|4 \\ d_2|3}} f(d_1)f(d_2) =$$

which it will be convenient for us to note is the same as

$$\left[\sum_{d_1|4} f(d_1) \right] \left[\sum_{d_2|3} f(d_2) \right] =$$

due to nothing fancier than distributing.

Example 3. Let n be a positive integer and f an arithmetic function. Prove

$$\sum_{d|n} f(d) = \sum_{d|n} f\left(\frac{n}{d}\right)$$

In particular this implies

$$\sigma(n) = \sum_{d|n} d =$$

Proof. Before we enter the proof, let's examine the two sides in the case $n = 12$.

$$\sum_{d|12} f(d) =$$

$$\sum_{d|12} f\left(\frac{12}{d}\right) =$$

This exposes the core idea.

List all the positive divisors of n in increasing order.

Next list n over each of these divisors.

This enumerates the divisors in the:

Therefore the two sums $\sum_{d|n} f(d)$ and $\sum_{d|n} f\left(\frac{n}{d}\right)$ only differ by

and hence the two sums are equal.

□